

HOUTRIB, Netherlands

WMO: 062680

Lat: 52.533N

Lon: 5.433E

Elev: 7

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.3	21.4	10.3	9.3	19.5	15.0	11.8	23.0	42.2	44.5	38.7	45.5	13.0	70	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	79.2	67.0	75.6	65.1	72.6	64.6	69.6	76.2	67.6	73.2	65.6	70.6	12.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	99.7	73.8	65.2	93.2	71.4	63.5	87.7	68.8	33.6	76.3	31.9	73.1	30.4	70.9	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.6	29.9	27.3	DB	16.9	85.7	8.5	4.2	10.8	88.7	5.9	91.2	1.1	93.5	-5.0	96.6	(4)
(5)			WB	16.1	73.3	8.5	2.0	10.0	74.7	5.0	75.9	0.3	77.0	-5.9	78.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	38.2	36.1	42.6	47.2	55.5	58.7	64.1	63.1	58.0	51.2	43.5	40.0
	DBStd	11.01	7.32	7.74	6.05	5.70	5.98	4.99	4.56	4.21	3.75	4.82	6.07	5.27
	HDD50	1683	365	391	236	121	12	1	0	0	0	45	201	311
	HDD65	5596	830	810	694	535	303	200	75	87	214	428	644	776
	CDD50	1660	0	1	7	37	182	262	436	407	239	82	6	1
	CDD65	96	0	0	0	0	8	11	46	29	2	0	0	0
	CDH74	474	0	0	0	6	69	54	207	125	13	0	0	0
	CDH80	77	0	0	0	0	5	5	34	31	2	0	0	0
Wind	WSAvg	14.4	16.6	15.5	15.7	14.3	14.2	13.5	13.6	13.5	13.1	14.1	13.7	15.4
Precipitation	PrecAvg	33.3	2.9	2.3	2.0	1.7	2.2	2.8	3.3	3.6	3.1	3.3	3.2	3.1
	PrecMax	45.0	5.6	5.4	4.0	3.7	3.8	6.0	6.1	8.1	7.0	8.3	5.9	5.9
	PrecMin	24.1	0.3	0.7	0.4	0.0	1.1	1.1	1.1	0.5	0.5	0.6	0.4	0.9
	PrecStd	4.8	1.3	1.1	1.1	0.9	0.8	1.3	1.5	2.0	1.6	1.6	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	54.4	61.7	71.8	80.7	79.5	84.8	84.1	74.5	66.8	56.4	53.2
		MCWB	48.3	51.7	53.4	57.6	63.3	66.8	71.1	70.7	65.7	60.5	52.3	49.8
	2%	DB	49.1	50.5	56.0	64.5	75.5	74.7	79.5	76.8	69.3	62.3	53.9	50.7
		MCWB	46.7	48.0	49.7	56.5	60.6	64.5	68.3	66.8	62.9	58.4	51.2	48.4
	5%	DB	47.3	47.1	52.5	59.6	71.0	71.5	75.8	72.9	66.1	59.8	52.4	48.9
		MCWB	45.3	44.7	48.7	54.1	59.5	64.0	67.2	65.7	61.6	56.4	50.4	46.8
	10%	DB	45.9	44.8	50.1	56.0	65.7	67.8	72.4	70.2	63.9	58.1	50.9	46.9
		MCWB	44.3	42.8	47.1	51.4	58.1	61.9	65.8	64.3	60.0	55.2	49.2	45.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	51.9	53.5	60.0	65.0	69.9	72.8	72.5	66.9	61.2	53.6	50.3
		MCDB	50.3	54.1	59.5	69.4	74.0	75.8	80.8	82.7	72.7	65.5	55.0	52.1
	2%	WB	46.9	48.1	51.2	56.8	62.8	67.0	70.4	68.9	64.2	58.8	52.0	48.6
		MCDB	48.2	49.9	55.2	62.8	72.5	72.5	77.5	75.2	67.6	60.8	53.3	50.3
	5%	WB	45.6	45.0	49.1	54.5	60.5	64.5	68.4	66.6	62.5	57.4	50.6	47.2
		MCDB	46.9	46.6	51.9	59.1	69.0	69.6	74.3	71.6	65.4	59.6	52.0	48.7
	10%	WB	44.5	42.9	47.4	52.0	58.4	61.8	66.2	64.9	60.7	55.7	49.3	45.4
		MCDB	45.8	44.4	49.6	56.1	65.3	67.3	71.1	69.3	63.6	57.6	50.6	46.6
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.8	8.9	10.4	11.5	9.7	10.1	10.4	9.0	9.2	6.7	5.8
		MCDBR	7.0	9.7	13.6	16.5	21.0	17.5	17.5	16.9	13.6	12.4	7.4	7.7
		MCWBR	6.5	7.9	9.4	10.3	10.5	10.7	10.2	9.6	8.9	8.8	6.4	7.0
	5% WB	MCDBR	6.8	9.3	11.9	14.8	17.6	15.4	15.2	14.7	11.8	11.5	7.0	7.5
		MCWBR	6.6	8.1	8.9	9.8	10.4	10.5	10.0	9.4	9.0	8.7	6.3	7.0
Clear-Sky Solar Irradiance	taub	0.339	0.359	0.390	0.412	0.414	0.411	0.423	0.415	0.392	0.373	0.351	0.333	
	taud	2.303	2.299	2.228	2.225	2.265	2.290	2.268	2.308	2.359	2.364	2.325	2.298	
	Ebn at Noon	201	234	250	260	266	267	261	257	248	226	194	180	
	Edh at Noon	20	27	35	40	40	40	40	36	31	25	20	17	
All-Sky Solar Radiation	RadAvg	213	431	826	1339	1639	1736	1656	1375	983	555	246	158	
	RadStd	21	76	89	125	99	108	165	108	82	49	23	17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page